

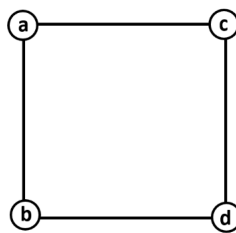
WORKSHEET 0: A FIRST LOOK AT GRAPHS

The research question we will be working on for the next four weeks focuses on a mathematical object called a *graph*. This is different from the graph you may already be familiar with, and we will define it now.

DEFINITION. In graph theory, the term **graph** refers to an object built from vertices and edges in the following way.

- A **vertex** in a graph is a node, often represented with a dot or a point. (Note that the singular form is vertex and the plural form is vertices.)
- The **edges** of a graph connect pairs of vertices. We usually represent the edges as straight or curved lines.
- In graph theory, a **graph** is a set of vertices and edges.
- Two vertices are **adjacent** if they are connected to each other by an edge.
- We often give the vertices labels (such as letters or names). Edges can be named by listing the two vertices the edge connects. For instance, the edge AB would connect two vertices labeled A and B.

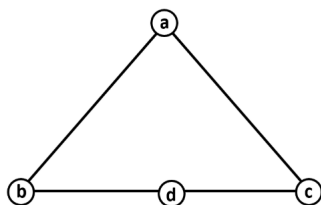
Example. Here is an example of a graph



The vertices are a , b , c , d .

The edges are ab , bd , cd , ac .

Exercise 1. Write down the vertices and edges for the following graph.



Remark. These two graphs are exactly the same graph! Since they both include the same four vertices connected in the same way, they represent the same graph. Because of this, graphs are useful when we want to model a situation for which connections are important and geometry is not.

DEFINITION. The **degree** of a vertex is the number of times it meets an edge. If a vertex is not connected to any edges, it has a degree of 0. See the diagram below for an illustration of the degrees of several vertices.

Degree 0	Degree 1	Degree 2	Degree 3	Degree 4

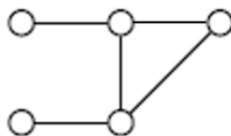
Exercise 2. Draw a graph that has exactly four vertices each of degree 3.

Exercise 3. Draw a graph that has exactly six vertices, half with degree 2 and half with degree 4.

Remark. Sometimes, graphs will have **loops**, which are edges that begin and end at the same vertex. Sometimes graphs allow **multiple edges** between the same two vertices. For our work, we will **not** allow our graphs to have either of these.

DEFINITION. A **path** is a sequence of distinct, adjacent edges. A path may be named by listing the vertices in the path in the order they are visited.

Exercise 4. Label the vertices of this graph and find a longest path.



DEFINITION. We say that a graph is connected if it's all one "piece". Using paths, we can describe more exactly what we mean by this.

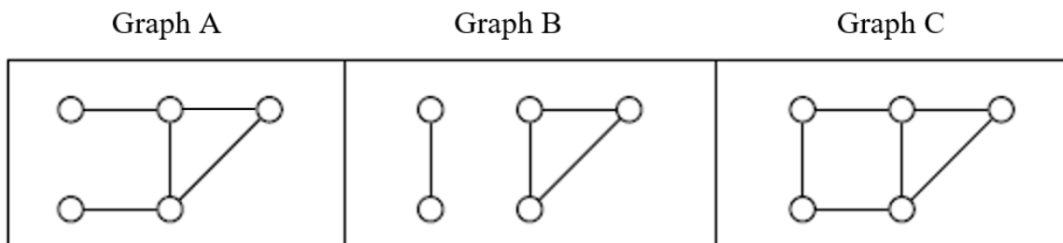
- A graph is called **connected** if there is a path between each pair of vertices.
- A graph that is not connected is called **disconnected**.
- The different "pieces" of a disconnected graph are called **components**. Each component is, itself, connected, while no component is connected to another component.
- A **bridge** in a graph is an edge whose removal would create a new component.

Exercise 5. Draw a graph that has four vertices each of degree 1.

Exercise 6. Can you draw a connected graph where each vertex has degree 1?

Exercise 7. Draw a graph with seven vertices, six edges, one bridge, and one component.

Exercise 8. Identify the bridges (if any) in each of the graphs.



Exercise 9. Is Graph B connected or disconnected? Justify using the definitions above.

Exercise 10. Draw a graph to fit each of the descriptions below. Label each vertex. Then for each graph, provide the following information: (i) next to each vertex, write its degree; (ii) give the total number of edges; (iii) give the total number of components.

- It has at least 7 vertices and exactly one bridge.
- It has at least six edges and at least two components.
- It has exactly four vertices each of degree 3.
- It is connected and has five vertices, exactly one of which has degree 1. Does your graph have any bridges?

e. It is connected and has five vertices, all of which have degree 2. Does this graph have any bridges?

Exercise 11. *Draw two different graphs, each with precisely 5 vertices and 6 edges. Are your graphs really different? How can you tell?*

Remark. *This is it for graphs for now! We will learn some more formal math to learn how to argue mathematical points to others. We will soon return to graphs once we are ready to introduce the research problem!*